

Literature Review Project

Global Land Temperature Analysis: A Data-Driven Approach
to Understanding Climate Change Patterns

Nouf Almojel	222410007
Leen alfawaz	222410838
Aljazy Altowayan	221410106
Noura alshnaife	221411094

Instructor: Dr. Heyfa Ammar

1) [Climate change: Global temperature](#)

R Lindsey, LA Dahlman

Paper Name: Climate Change: Temperature around the Globe

Summary: This paper gives an illustrative presentation of global temperature trends and explains the important features of the temperature increase that is caused by climate change. It reminds citizens of the need to better appreciate and yet explain variability within the climate and its consequences towards natural and man made systems. Using graphical analysis as its primary tool, the paper urges world leaders to launch a coordinated campaign against global climate change and increase peoples' awareness of the irreversible consequences of global warming with regard to ecosystems and populations.

2) [The relation between climate change in the Mediterranean region and global warming](#)

P Lionello, L Scarascia

Paper Name: Climate Change in Mediterranean and Global Warming – A Comparative Study

Summary: This paper is restricted to particular correlation between climate change in the Mediterranean and global warming especially in relation to the changes in the global annual mean surface temperature. The authors examine if and how this peculiar area of the world, characterized by extreme climate, is influenced by global temperature changes. The conclusions indicate that the conditions in the Mediterranean may reflect conditions in other parts of global warming generally, and the paper underlines the significance of regional climate research data for studying world climate change in particular.

3) [Future extreme climate changes linked to global warming intensity](#)

X Wang, D Jiang, X Lang

Paper Name: Impacts of Climate Change on Extreme Climate Changes in the Future: Relation to Intensity of Global Warming

Summary: This paper looks at how extreme future climate change is related to the global warming intensity factor. It implies that the Clausius-Clapeyron relation serves as a useful diagnostic, especially in relating heavy precipitation extremes to global temperature change. This paper brings empirical indications of a nonlinear association of warming with precipitation events, which might be useful for the prognosis and coping with the effects of severe weather occurrences. In this sense, by pinpointing this particular climate change factor, the paper gives a more correct understanding of how the global warming might worsen extreme weather events, with relation to the ones connected with precipitation in particular.

4) [Global warming, climate change, and environmental pollution: recipe for a multifactorial stress combination disaster](#)

SI Zandalinas, FB Fritschi, R Mittler

Paper Name: Global Warming, Climate Change and Environmental Pollution: An Anticipated Stress Multi factorial Stressor Combination Disaster

Summary: Indeed this paper looks at global warming, climate change and environmental pollution as a stress combination factor Disaster. The authors explain global temperature anomaly as temperature variation from a base temperature and associate it with negative consequences on natural and human systems. In this study, the authors also argue that the disasters may be more frequent due to pollution and climate change, especially if the level of temperature continues to rise. Therefore, this analysis acknowledges the need userList that several environmental issues should be managed concurrently.

5) [Strategies for mitigation of climate change: a review](#)

S Fawzy, AI Osman, J Doran

Paper Name: An overview and Critical Analysis of Climate Change Control Measures

Summary: This review paper offers a comprehensive overview of the many approaches that are advocated to address the climate change problem and the primary goal of preventing global temperatures from rising beyond an increase of 2°C above pre-industrial levels. The authors look at the options for a world transitioning to renewables, carbon capture, and changing land use to provide a framework for bringing

about climate stabilization. From the paper, it is clear that the author supports the use of the multiple strategies as key in achieving the intended international climate aim. Lessons learnt have been indicated to present challenges or trade offs but at the same time the paper affirms the need for specific actions to be taken.

6) [The human imperative of stabilizing global climate change at 1.5 C](#)

O Hoegh-Guldberg, D Jacob, M Taylor...

Paper Name: Climate Change 2014: Mitigation of Climate Change: Working Group III – <http://report.mitigation2014.org> & Working Group III Assessment 2014 Report.

Summary: Hence this paper will discuss the imminence of stabilizing climate change at 1.5°C concentrating on impacts likely to affect human communities especially those in the agriculture sector. It shows how temperature increase leads to relocation from agri-villages; a feature that raises the alarming possibilities of climate change affecting human behavior. This paper weighs the need to limit global warming to 1.5°C which it states is the buffer that will prevent large-scale displacement and maintain social order. It urges the U.N to address ways in which it will ensure policies support affected groups of people in combating the impacts of climate change.

7) [Drylands face potential threat under 2 C global warming target](#)

J Huang, H Yu, A Dai

Paper Name: Drylands at Risk as Warming Limit Sets by Paris Accord Stands at 2°C

Summary: This paper focuses on the ability of dryland ecosystems to adapt to a circumstance where the global warming is maintained at a 2°C regime: this paper notes that a global warming regime of 1.5 °C is important to avoid damage to the ecosystems. The authors of the paper discuss different effects of temperature rise including desertization, agricultural yields decline and biodiversity loss, on the drylands. As illustrated by the drylands' example, the paper reveals the overall impact of climate change on fragile biogeographic provinces. This horn calls for solidification of climate policies in protection of these areas and equally emphasizes the symbiotic relationship between ecology and climate.

Comparison of the 7 Papers

Paper	Advantages	Limitations
1	A very relevant and helpful view of global temperature trends and climate change, in general.	Does not contain clear elaboration and consideration
2	Selective based on stress on climatic change in Mediterranean region and global warming	However, some of the results derived in this study are region specific; therefore, it may not apply to other regions of the world countries formed part of the study.
3	Investigates the relationship between further extreme changes in climate in the future and increasing global warming intensity, clausius-clapeyron equation	Concentrates with one particular relationship and likely to leave several aspects of global temperature changes untouched
4	Addresses the compound stress interact disaster arising from global warming, climate change, and environmental pollution	Explains what global temperature anomaly is, but doesn't offer much exploration on this subject.
5	Summarizes the key approaches to approach climate change with the overall target of preventing global mean temperature rise to more than 2 degrees Celsius.	Isn't as comprehensive as it discusses mostly the measures which will be taken in order to avoid further increase of the global temperature.
6	Talks about human endeavor to make global climate change to be 1.5°C, realizes that out-migration and global temperature are positively related.	Possibly the human effect of global temperatures might not address more general impacts on the environment.
7	Examines the possible vulnerability of drylands when the global temperature rise remains at 2°C while stressing that it is necessary to achieve a 1.5°C temperature rise limit.	Concentrates on one ecosystem (drylands), seems to miss other aspects of global temperature alterations

Suggestions for Improvement:

After analyzing the seven papers, several key areas for improvement emerge:

- **Integration of Multiple Data Sources:** The majority of researchers carry out their findings with a small sample of data. We propose integration of several temperature datasets, such as satellite ones, ground-based measurements, and ocean data sets to develop a much more comprehensive analysis system.
- **Advanced Analytical Methods:** Research done today utilizes old school statistical analysis methods. It is therefore suggested that application of more sophisticated ANNs and ensemble methods should help to improve the capture of the underlying temperature patterns.
- **Real-time Analysis Framework:** A large number of investigations are limited to the descriptive approach only. As a recommendation, we refer to the creation of a real-time system for temperature forecasting that would imply state-of-the-art sensor networks and automated data acquisition.
- **Multi-scale Analysis:** Mainly the current research is inclined towards dealing with both global and regional scales, but in isolation from one another. The idea of creating a single framework for analysing patterns across various spatial scales is also suggested, but with an added respect for local climate.
- **Improved Uncertainty Analysis:** Sometimes, you will find that existing models represent very wide confidence intervals. To address these issues, we propose the use of improved methods of uncertainty quantification, and probabilistic forecasting.

Contribution and Project Overview:

Our project makes several significant contributions to global temperature analysis:

- **Enhanced Data Processing:** It is for these reasons that we have constructed a completely holistic preprocessing framework in which we have high data retention rates due to missing value imputation but which still retains reasonable statistical validity.
- **Historical Trend Analysis:** Our work is a comprehensive analysis of the time series temperature data and is useful for understanding long-term climate behavior.
- **Regional Pattern Identification:** We have worked out strategies to detect relevant temperature changes and recognize regions that should be addressed and treated as a priority.
- **Seasonal Variation Analysis:** By suitably capturing and quantifying seasonal temperature differences, our proposed framework is useful in understanding the climate interface and developing climate change adaptation models.
- **Predictive Modeling Framework:** We organized temperature forecasting based on the concept of volatility and incorporated the conditional volatility into our forecast model; therefore, the models serve as reference for both research and policy planning.

This project fulfills the gap between the historical evaluation and the prediction part and at the same time making sure that the global and regional perspectives are balanced nicely. Besides, it not only offers scientific contributions to discuss the global temperature changes but also offers specific resources and knowledge to the climate policy makers and researchers. The work can act

as a climate change bibliographic database, a tool for analysis and a base for proposing hypotheses and future research and policy endeavors.